

03-09-2023

PHYSICS Section A

Ans. 1. 4) presbyopia

Ans. 2. 2) repulsion

Ans. 3. 4) virtual, erect, same size as you

Ans. 4. 2) below image of fish
Section B

Ans. 5. given,
 $P = +1.5D$

$$P = \frac{1}{f}$$

$$1.5 = \frac{1}{f}$$

$$f = \frac{+100}{+1.5 \times 3} = 0.67m = \boxed{67cm}$$

$\therefore$ focal length is 67cm and it is converging as its focal length is positive

Ans. 6. given,

$$\text{time} = 30 \text{ min} = 1800 \text{ sec}$$

$$\text{current flowing} = 5A$$

$$I = \frac{Q}{t}$$

$$Q = It$$

$$Q = 5 \times 1800$$

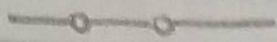
$$Q = \boxed{9000C}$$

$\therefore$ charge circulated is 9000C.

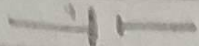
Ans. 7.



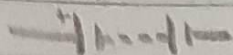
- Switch open



- Switch closed



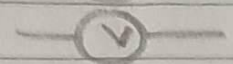
- cell



- battery



- ammeter



- voltmeter

Section C

Ans. 8.

Given,

$$f = 12 \text{ cm}$$

$$h = 5 \text{ cm}$$

$$u = -20 \text{ cm},$$

Using lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{12} = \frac{1}{v} + \frac{1}{20}$$

$$\frac{1}{v} = \frac{1}{12} - \frac{1}{20}$$

$$\frac{1}{v} = \frac{5-3}{60} = \frac{2}{60}$$

$$v = 30 \text{ cm}$$

$$\frac{v}{u} = \frac{h'}{h} \quad (\text{magnification formula})$$

$$\frac{30}{-20} = \frac{h'}{5}$$

$$m = -\frac{3}{2} = \frac{h'}{5}$$

$$h' = -\frac{15}{2}$$

$$h' = -7.5$$

Image is of 7.5cm, enlarged, real, inverted and at distance of 30cm from the lens.

Ans 9. Given,
electric potential 1st end = 10V
" 2nd end = 5V

work done = 5 J

Now,

work done = $Q(\Delta V)$

$$W = Q \times \Delta V$$

$$5 = Q \times -5$$

$$[\Delta V = 5 - 10 = -5V]$$

$$Q = -1 \text{ Coulomb}$$

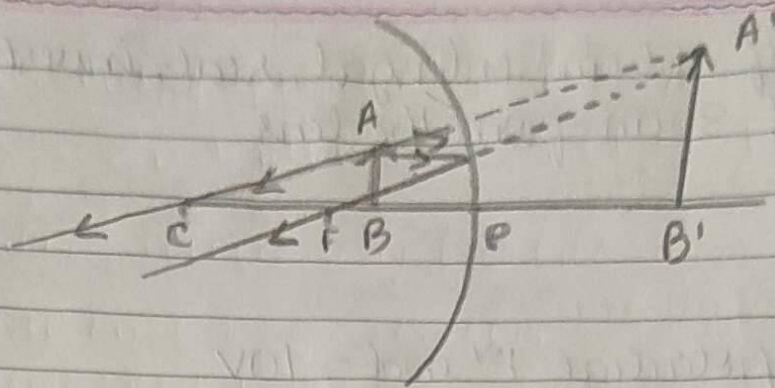
Section D

- Ans 10.
- Pole is center of the spherical mirror, represented by P.
 - Radius of curvature is the radius of the sphere which the mirror is a part of.
 - Principal axis is a straight horizontal line passing through pole and center of curvature of concave mirror.
 - Principal focus is point at which ray parallel to principal axis after being reflected from concave mirrors are met.

To obtain an erect image, the candle flame should be placed at distance of 0-20cm from the mirror.

Properties $\rightarrow$

- image is behind mirror
- image is enlarged
- image is virtual.



Ans. 11.

- a) Red light scatters the least by molecules present in the air. During early morning, sunlight travels a longer path through atmosphere to reach our eyes. Blue light scatters most and has been removed, leaving red light remaining which then reaches our eyes. Hence it appears reddish.
- b) Power of accommodation of eye is its ability to focus near and far objects clearly on retina by ~~the~~ adjusting the focal length of ~~the~~ eye lens.
- c) Least distance of distinct vision of a normal human eye is 25 cm.